☆ Assignment - 1 Date: 09/02/2026

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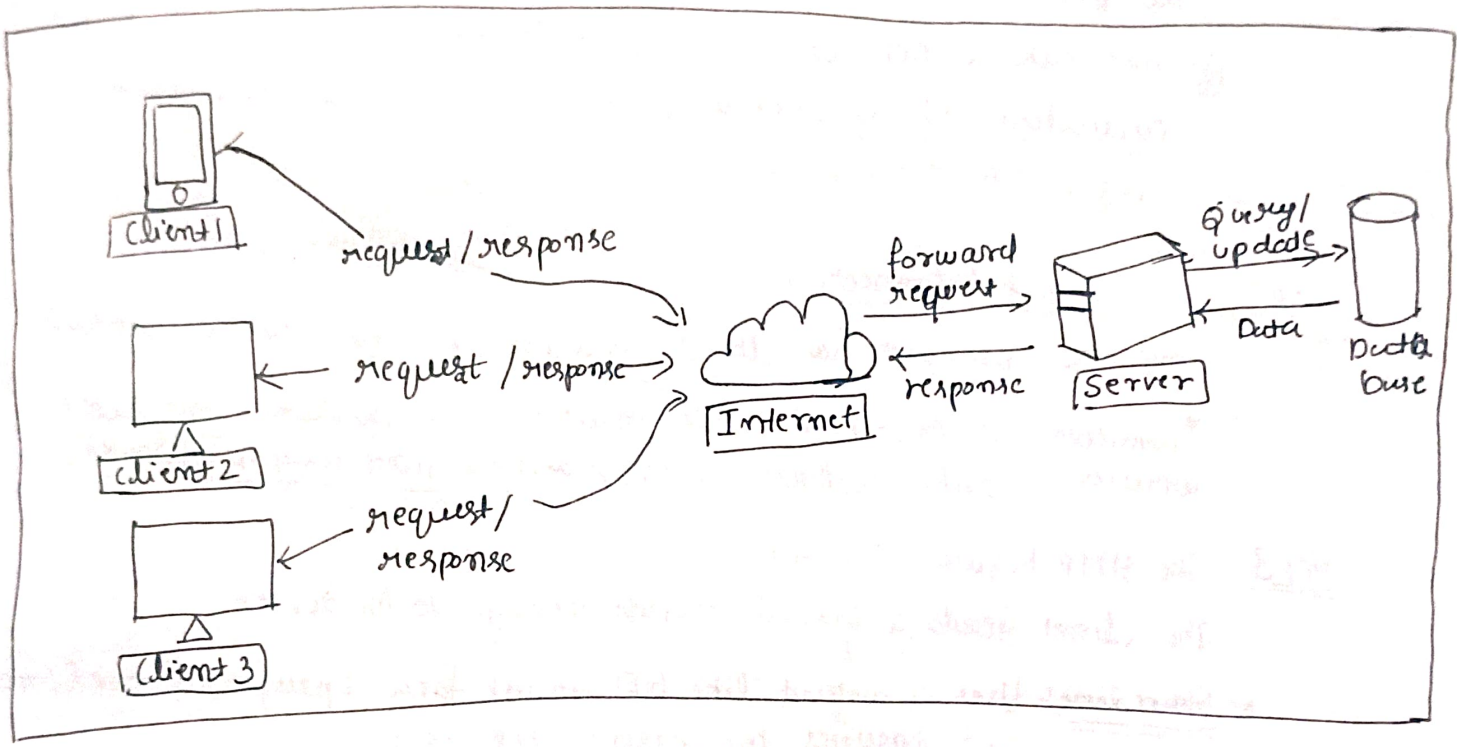
☆ Name: Internet explorer

☆ Description:-

Draw and Explain the Client Server architecture and trace what happens when you open a website.



CLIENT- SERVER ARCHITECTURE

☆ The Clients:- Devices like Smartphones or Computers that initiate a Request for information or Services.

☆ The Internet:- The global network that acts as a bridge carry data between the client and the Server.

☆ The Server:- A Control Computer that receives requests, runs the necessary logic, and provides the appropriate response.

☆ The Database:- A dedicated storage System where data is organized, retrieved and updated by the Server.

Tracing a website Request

Tracing a web site Request

when we type www.google.com into your browser & hit Enter, a lot happens in the blink of an eye:

Step 1:- DNS Lookup – Finding the address

The browser takes the domain name (www.google.com) and asks a DNS Server → Domain Name System → "Phone book" equivalent of ip addresses, for its numerical IP Address (eg., 216.58.211.206).

Step 2:- TCP/IP Socket Connection :-

once the browser has the IP Address, it establishes a virtual connection to the server, This ensures that the data sent later arrives in order without errors. → using a Three way handshake.

Step 3:- The HTTP Request (The Ask)

The client sends a formal request message to the server

- **Start line:-** Uses a method like GET to ask for a specific resource (/maps) and specifies the version (HTTP/1.1).
- **Headers :-** Extra information, such as what type of language the user prefers or what browser they are using.
- **Body** → usually empty for simple page views, but used if you are sending data (like a login/Password) to the server.

Step 4:- The HTTP Response :-

The server processes the request & sends a back a response:

- **Status Code:** A 3 digit number telling the browser if it was successful. 200 OK means it worked. → or it was failure → 404 Not found
- **Response headers :-** information about the data being sent back, like the data & the type of content.
- **Response body :-** The actual "Content" of the Response – the HTML code that your browser turns into the website you see.